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US Permanent Resident

Youzhi Wang

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Education

Columbia University
MS Computer Science
GPA: 3.96/4.0

New York, NY
Dec 2025

University of California, Santa Barbara
BS (double major) Applied Mathematics, Statistics and Data science
Graduated with Honors and Major(Applied Math) distinction

Santa Barbara, CA
June 2023

Skills

Programming Languages: Python, Java, Bash, C/C++, R
Tools/Frameworks: PyTorch, sk-learn, SQL, AWS, Docker, Linux, Git, Vim, Tmux, L^AT_EX

Projects

Full Stack Project: Second-Hand Product Trading Site (New York, NY) **Sep 2024 – Current**

- Applied **Agile** methodology with two-week sprints and the **Kanban** model to streamline project workflow, while using the **RACI** framework to clearly define team roles and responsibilities
- Deployed 3 microservices on 3 separate **Amazon Linux 2 (AWS EC2)** instances using **Python Flask** as backend and **React.js** as frontend UI
- Used **Docker** to containerize order tracking microservices for environment consistency
- Connected each microservice to a separate **MySQL** database to prevent dependencies and minimize downtime
- Coordinated version control for 7 member team using GitHub

Schizophrenia Classification Using Deep Learning (New York, NY) **Jan 2024 – May 2024**

- Trained Neural Networks on 3D MRI brain images that achieved AROC of 0.87 using **PyTorch**
- Visualized regions of the brain that cause schizophrenia using **Python** and **JupyterNotebooks**
- Compared the performance of 2D vision transformers and 3D Convolutional Neural Networks using **Python scripts** and found that 3D CNN models still performs better for brain MRIs

Multiple Sclerosis Classification Using Deep Learning (New York, NY) **Sep 2023 – Jan 2024**

- Created **BASH** scripts to segmented 100 brains from MRI images and extracted Radiomics features
- Developed **Python** scripts with **PyTorch** for Machine Learning training and scheduled them to run on cluster computers through SSH

Earthquake Data Collection And Classification (Santa Barbara, CA) **Jul 2022 – Jun 2023**

- Collected and cleaned large collections of seismic signals (over 100,000 waveforms) using **Python** and ObsPy
- Improved signal identification time by 5 times through restructuring previous data processing pipeline using **Python**
- Implemented 5 different **Machine Learning algorithms** (KNN, LDA/QDA, SVMs and DCNN) for classifying high quality seismic wave signals in **Python**

HTTP 1.0 Web Server **Sep 2023 – Dec 2023**

- Wrote **HTTP 1.0** web server using **C** and socket APIs with image hosting and interactive search module using linked list to search through given database

Java Command Line Program **Nov 2023 – Dec 2023**

- Created command line **Java** application to find all anagrams of given word from large text documents using Binary Search Trees, AVL Trees and HashMaps

Research Publication

VarBench: Robust Language Model Benchmarking Through Dynamic Variable Perturbation
(Accepted by EMNLP 2024) **June 2024**

- Enhanced a math reasoning benchmark for LLMs by developing over **400 Python functions** to generate math problems that assess reasoning capabilities.
- Created **Python scripts** to build comprehensive testing datasets for large language models.